tripleten

# Automation of Weather-Based Sales and Staffing Decisions

Raisa Vigo Fonseca

AI Automation | Project 5

# Problem

- Bayside Brews & Scoops needs an automated workflow to replace manual weather checks. Each morning at 7 AM, it should pull Miami’s forecast, use AI to recommend product mix, staffing, and two promo messages, then email it to manager Emily Davis. Severe weather should trigger an additional alert for quick preparation.

# Solution

## Automation Goal

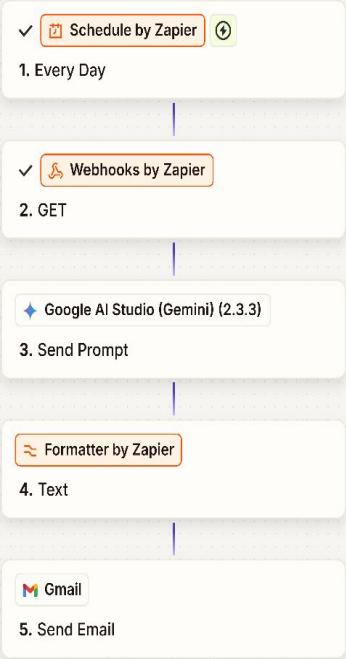
To automatically send a daily, weather-based business recommendation email that helps Bayside Brews & Scoops plan product mix, staffing, and promotions—eliminating manual weather checks.

## What the Workflow Does

- Runs every morning at 7:00 AM
- Pulls the Miami, FL forecast from OpenWeather
- Sends the forecast to Gemini to interpret conditions
- Generates product suggestions, staffing advice, and promo messages
- Delivers a clear summary email to manager Emily Davis

This automation enables faster preparation and smarter daily decision-making.

# Workflow diagram



# Workflow in action

## Daily Plan – Bayside Brews & Scoops (Miami, FL)

**Zapier Vigo** <vigosupply@gmail.com>
to rvigofonseca67

4:04 PM (29 minutes ago)

Good morning, Emily 👋 Here's today's snapshot for Bayside Brews & Scoops. Weather: 80.53 °F — clear sky.

Here are your daily operations recommendations for Bayside Brews & Scoops:

**Product Focus:** Given the 80.53°F clear sky in Miami, strongly focus on **cold drinks** and **gelato**. Hot drinks and pastries will likely see reduced demand today.

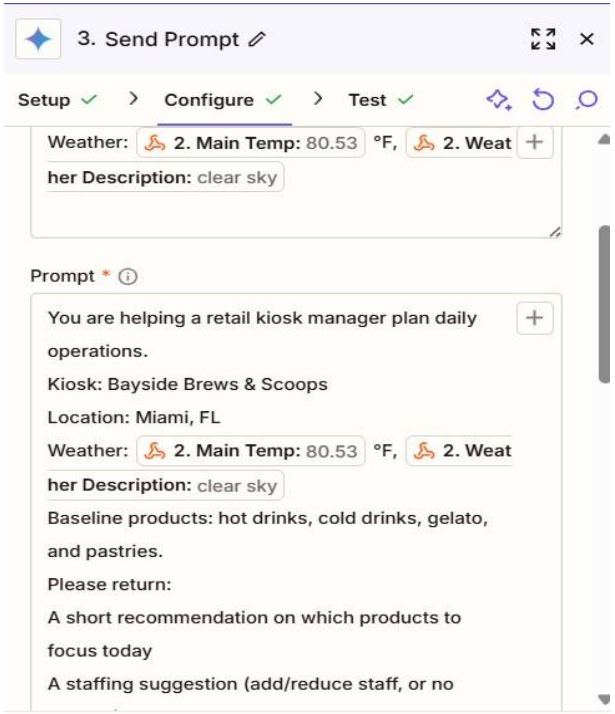
**Staffing Suggestion: No change**, but ensure staff are prepared for a potentially busy day with high demand for cold items. Consider extending shifts for key personnel if you anticipate peak crowds.

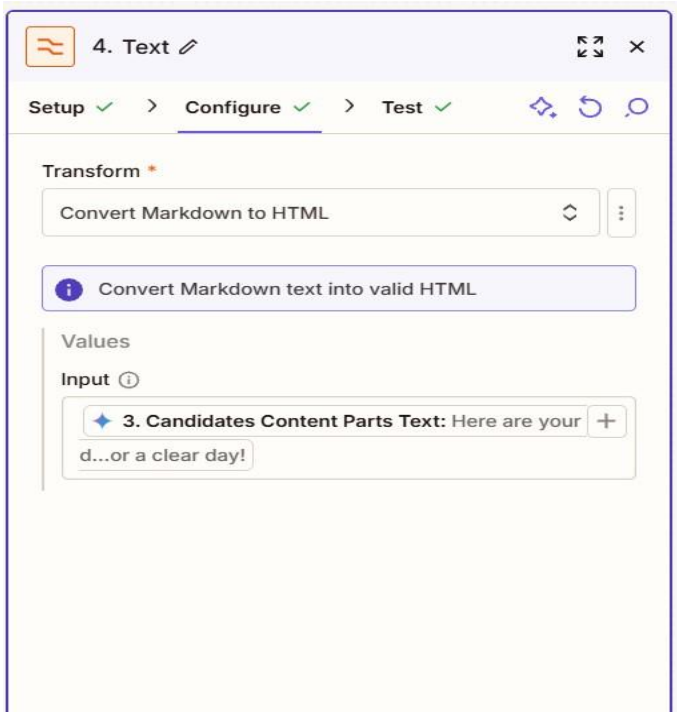
**Promotional Messages:** 1. Beat the Miami heat! Grab a refreshing cold brew or a scoop of our creamy gelato today! ☀️🍦 2. Sunshine & smiles! Cool down with our icy drinks and delicious gelato. Perfect for a clear day!

Generated automatically at 7 AM with weather + AI insights.

Reply Forward

# Workflow in action(2)



# Benefits

### 1. Faster, Data-Driven Decisions

The kiosk manager no longer needs to manually check weather apps or guess demand. The workflow delivers accurate, AI-generated recommendations every morning, improving product planning and staffing.

### 2. Increased Operational Efficiency

Automating weather retrieval, analysis, and email communication saves time and reduces human error. The team can prepare ingredients, inventory, and schedules earlier in the day with more confidence.

### 3. Higher Sales Through Weather-Based Promotions

By tailoring product mix and customer promo messages to the daily forecast, the kiosk can better match what customers want—boosting sales of hot drinks, cold drinks, or gelato depending on conditions.

tripleten

# Challenges & improvements

## What Worked Well

1. **Smooth Automation Flow**

Once connected, the Zapier steps ran in the correct order — scheduling, pulling weather data, sending the prompt to Gemini, formatting, and emailing. The workflow logic itself worked as planned.

2. **Reliable Weather Retrieval Using OpenWeather**

Using Webhooks + OpenWeather gave you consistent daily forecast data, which made the automation predictable and stable.

3. **Helpful AI Insights**

Gemini generated clear product and staffing suggestions, turning raw weather data into useful business recommendations. This added real value to the final email.

tripleten

# Challenges & improvements (2)

## What I'd Improve With More Time or Tools

1. **API Setup Was Challenging**

Configuring the OpenWeather API and structuring the query parameters took time.
Improvement:
Create a reusable API template or switch to a Zapier-integrated weather app with easier configuration.

2. **AI Output Formatting Issues**

Gemini sometimes returned inconsistent or unstructured answers. You had to add a Formatter step to "fix" the text.
Improvement:
Use a structured JSON prompt so Gemini always returns predictable fields, removing the need for heavy formatting.

3. **Better Error Handling**

Right now, if the API fails or the AI output is incomplete, the workflow may still continue.
Improvement:
Add filters or Paths in Zapier for: Missing weather data,AI output that doesn't contain required fields
This would make the workflow more reliable.